BRIGHTLEARN • DATA ANALYTICS CASE STUDY

Bright Motors

Car Sales Analysis

Business Insights for the New Head of Sales

By Lutendo Lucky Pau

Junior Data Analyst

Dataset: Bright Car Sales • 548,400 transactions • Jan 2014 – Jul 2015

OVERVIEW

Agenda

01 **Executive Summary**
Headline KPIs and key takeaways

02 **Methodology**
Data flow, cleaning and tooling

03 **Revenue Drivers**
Top makes, models and body types

04 **Pricing Dynamics**
Year of manufacture and mileage impact

05 **Regional Performance**
Where Bright Motors should focus

06 **Customer Trends**
Colors, transmission and margin tiers

07 **Recommendations**
Actions to grow profit and efficiency

AT A GLANCE

Executive Summary

R138.88B	548,400	R253,252	973
Total Revenue	Vehicles Sold	Avg Selling Price	Unique Models

Key Takeaways

- Ford, Chevrolet and Nissan lead revenue — top 3 makes drive a major share of 207,696 transactions.
- SUVs and Sedans dominate body-type revenue — clear inventory focus areas.
- Average margin sits at -15.0% — many sales below MMR. Pricing discipline is the biggest profit lever.
- Top 3 states (CA, FL, TX) generate the bulk of revenue — expansion should reinforce these hubs first.
- Newer cars (2013–2015) command 2–3x the price of 2005-era stock; mileage <50k miles holds premium value.

HOW WE GOT HERE

Methodology & Data Pipeline

1 Source
CSV file
548,400 rows
16 columns

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2 ETL
Clean nulls
Numeric prices
Date parsing

›

3 Storage
SQL warehouse
(Snowflake)
Staged tables

›

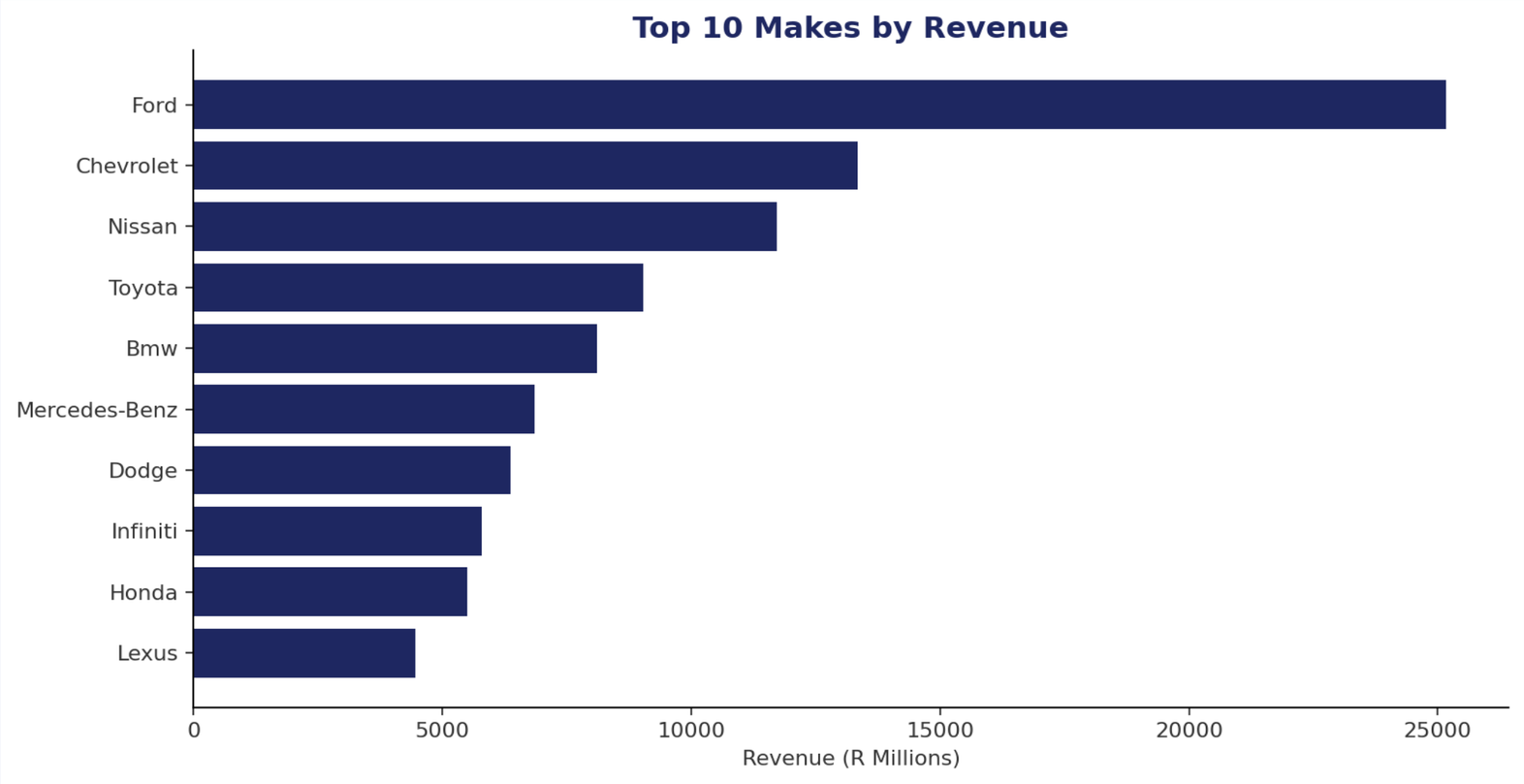
4 Analysis
SQL aggregations
Python / Pandas
Margin tiers

›

5 Visualize
Matplotlib
Dashboards
This deck

Calculations: total_revenue = selling_price × units_sold | margin% = (price − MMR) / price × 100

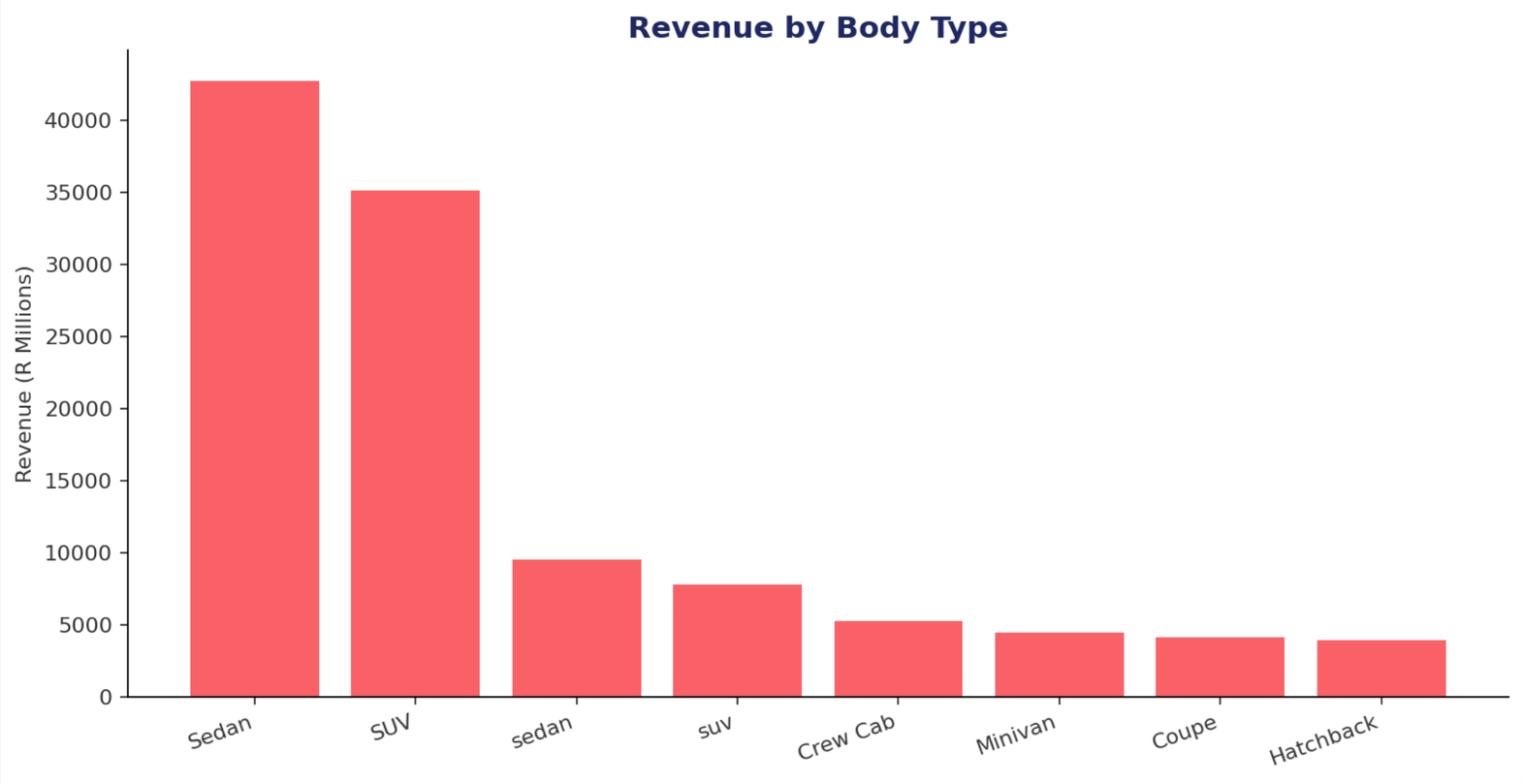
Revenue by Make



Insights

- Ford leads with R25,166M across 93,553 units.
- Chevrolet and Nissan round out the top 3 — combined they generate the majority of dealership revenue.
- Average price across the top 10 makes: R288,937.
- Action: Secure preferred supplier deals with the top 3 makes to protect inventory and margin.

Revenue by Body Type



Insights

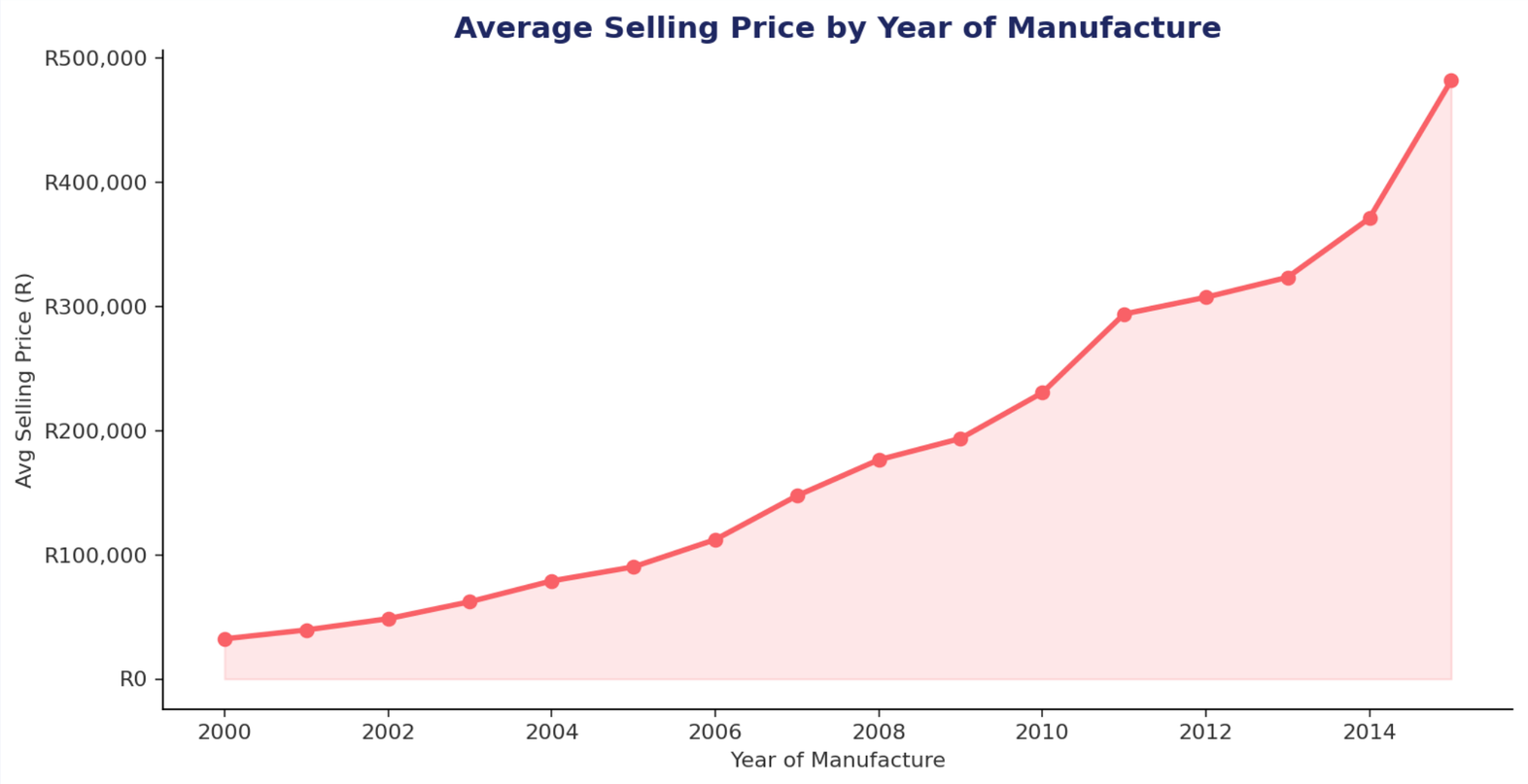
- Sedan dominates with R42,685M in sales.
- SUVs and Sedans together represent the strongest categories — both should be over-indexed in inventory.
- Specialty bodies (Coupe, Convertible) move lower volume but at higher average prices.
- Action: Maintain depth in SUV/Sedan stock and use specialty cars as high-margin showcase units.

Top 10 Models by Revenue

#	Make	Model	Units Sold	Revenue	Avg Price
1	Ford	F-150	14,479	R5,044.4M	R348,394
2	Nissan	Altima	19,349	R4,088.2M	R211,290
3	Ford	Escape	11,861	R3,068.9M	R258,736
4	Ford	Fusion	12,945	R2,958.4M	R228,535
5	Infiniti	G Sedan	7,417	R2,736.5M	R368,949
6	Ford	Explorer	7,707	R2,596.3M	R336,876
7	Toyota	Camry	12,545	R2,594.6M	R206,821
8	Bmw	3 Series	8,204	R2,551.3M	R310,978
9	Ford	Edge	5,915	R2,051.1M	R346,754
10	Chevrolet	Silverado 1500	6,362	R1,985.4M	R312,071

Insight: A small set of nameplates concentrates revenue — focused marketing on these models compounds returns.

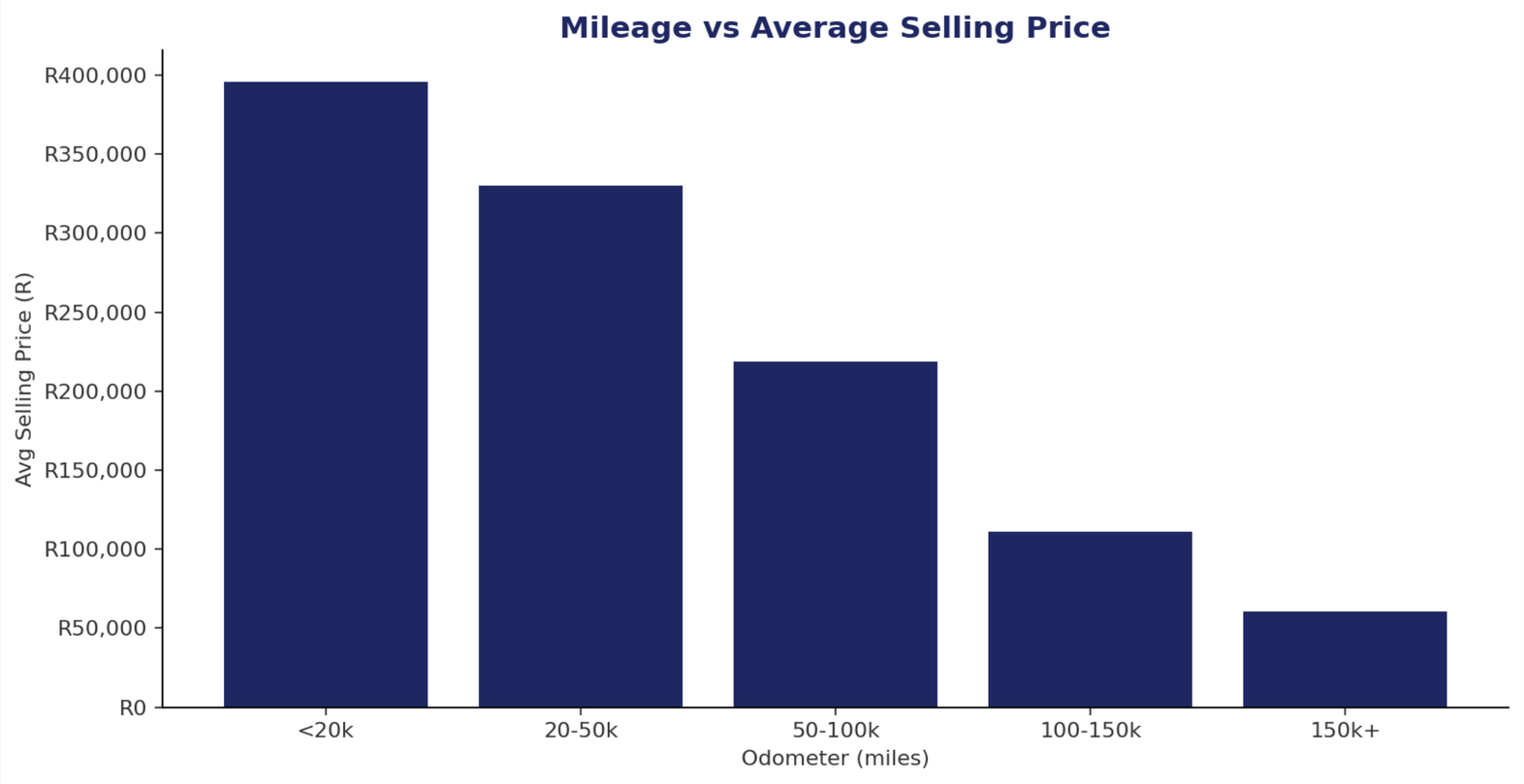
Year of Manufacture vs Price



Insights

- Strong, consistent price premium for newer model years.
- 2013–2015 vehicles command 2–3x the price of 2005 stock.
- Pre-2008 inventory underperforms; consider auction off-loading.
- Action: Bias purchasing toward vehicles ≤3 years old to maximise revenue per unit.

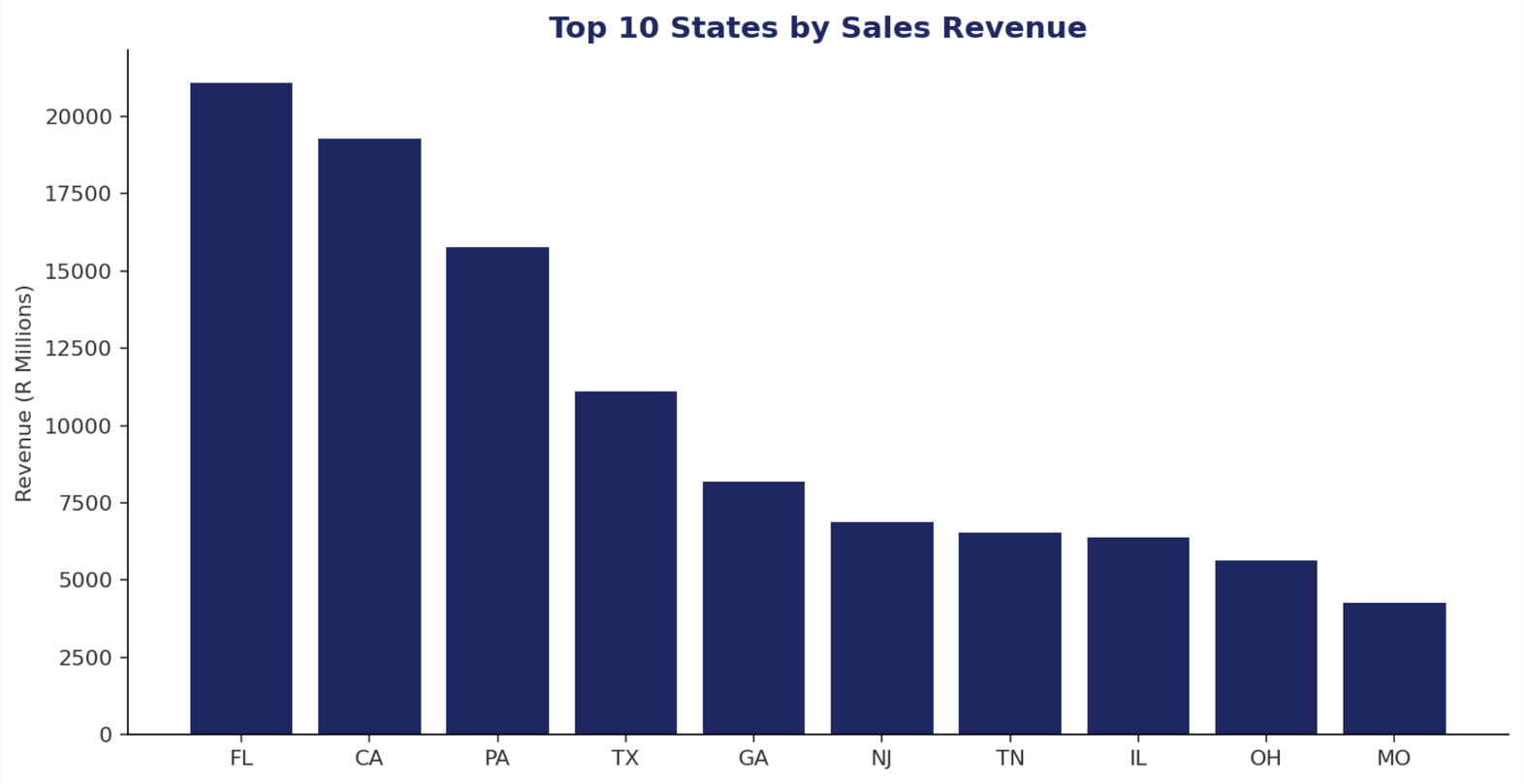
Mileage vs Average Price



Insights

- Cars under 20k miles fetch the highest average prices.
- Sharp price drop appears between 50k and 100k miles.
- Vehicles >150k miles convert at the lowest prices — limited upside.
- Action: Prioritise low-mileage trade-ins; price aggressively above 100k miles to keep stock turning.

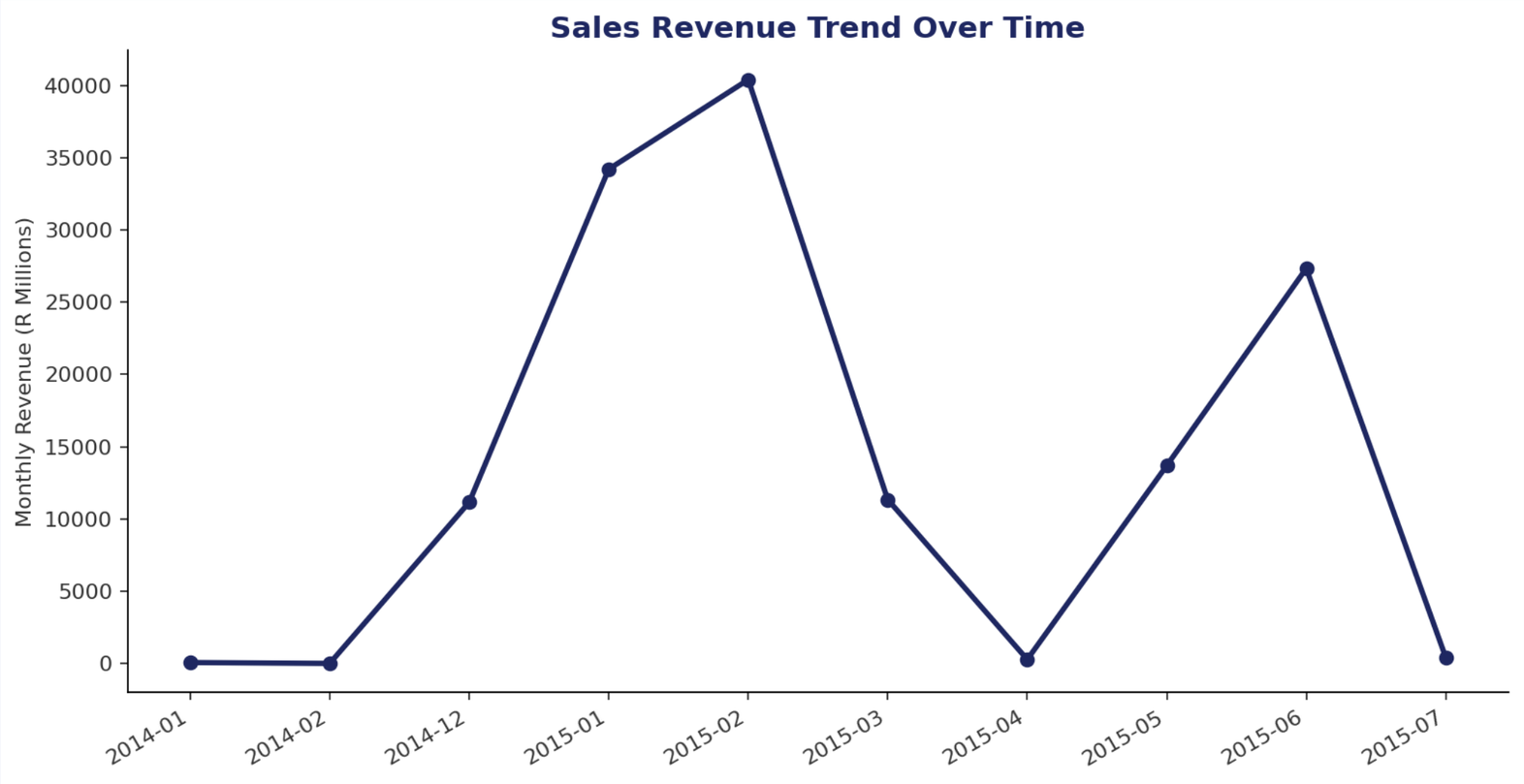
Top States by Revenue



Insights

- FL is the #1 market — R21,068M revenue.
- Top 3 states deliver disproportionate revenue vs the long tail.
- Untapped states represent expansion opportunities for the new dealership network.
- Action: Open 2 new dealerships in top-performing metros; pilot online sales for low-volume states.

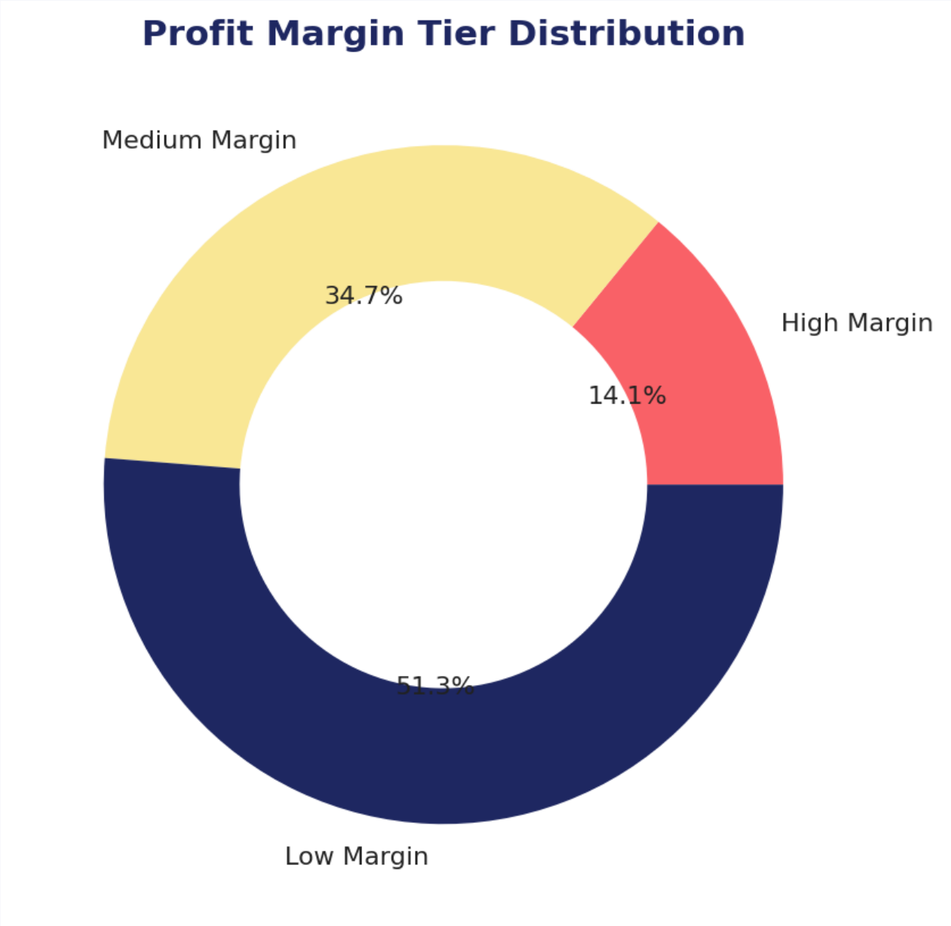
Sales Revenue Trend

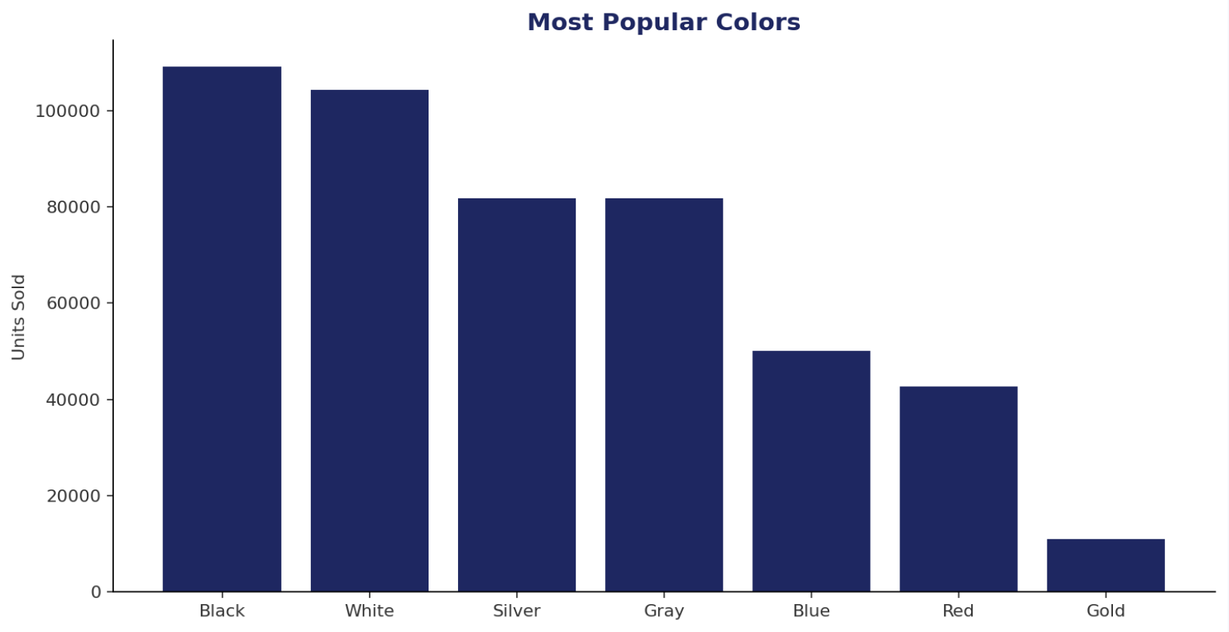


Insights

- Revenue shows clear seasonality with Q4 / Q1 strength.
- Steady volume month-over-month indicates a healthy demand baseline.
- Inventory planning should align purchases with seasonal demand peaks.
- Action: Build dynamic forecasting model to size inventory each month.

Profit Margin & Customer Preferences





Insight: Black and White account for ~40% of all units — stocking these colors hedges turnover risk. Over half of sales fall into Low-Margin tier; pricing tightening here is the single biggest profit unlock.

Recommendations to the Head of Sales

1. **Tighten pricing discipline**
   Half of all sales close below MMR. Implement minimum-margin guardrails per tier to lift average margin from −15% toward +5%.

2. **Concentrate inventory on winners**
   Over-index SUVs and Sedans from Ford, Chevrolet and Nissan — proven revenue drivers across regions.

3. **Target newer, low-mileage stock**
   Vehicles ≤3 years old and <50k miles command consistent price premiums. Bias trade-in offers accordingly.

4. **Expand dealership footprint in top states**
   Open new dealerships in CA, FL and TX metros where demand is proven; pilot e-commerce for outlying states.

5. **Build a monthly forecasting dashboard**
   Use the cleaned dataset in Power BI to forecast units, revenue and margin — refreshed weekly for the sales leadership team.

Thank You

Questions & Discussion

Lutendo Lucky Pau
Junior Data Analyst • Bright Motors Case Study
Prepared for the Head of Sales